

Batch Production Record - Selective Laser Melting of 316L Stainless Steel Flat Washers

Document ID: BPR-SLM-316L-2023-10-15-001

Revision: 1.0

Date: October 15, 2023

Prepared By: John Doe, Production Supervisor

Approved By: Jane Smith, Quality Manager

1.0 Batch Information

Parameter	Details
Batch Number	SLM-316L-WSH-231015
Product Description	316L Stainless Steel Flat Washers
Target Quantity	33 pieces
Equipment ID	SLM-Machine-05
Build Platform ID	BP-316L-89
Production Date	October 15, 2023
Shift	Day Shift (07:00-19:00)
Lead Operator	Michael Chen
Quality Inspector	Sarah Rodriguez

2.0 Material Preparation and Verification

07:15 - Material preparation initiated by Operator Michael Chen.

Raw materials verified against specifications and logged into system.

Material	Lot Number	Quantity	Verification Status
X2CrNiMo1712 Stainless Steel (316L base)	LOT-SS-2309-A	4.11 kg	Approved
Process Water (Atomization Grade)	N/A (Plant Supply)	16.8 kg	Approved

07:45 - Water atomization process commenced for powder production.

The water atomization electricity consumption was calculated based on the established energy requirement of 2.23 MJ per kilogram of material processed. For the 4.11 kg of stainless steel, this resulted in 2.55 kWh of low-voltage electricity usage for the atomization process.

08:30 - Powder quality inspection completed. All parameters within specification.

Parameter	Result	Specification
Particle Size Distribution	15-45 μm	15-53 μm
Apparent Density	4.1 g/cm ³	4.0-4.3 g/cm ³
Flow Rate	25 s/50g	20-30 s/50g

3.0 SLM Process Execution

09:00 - SLM machine preparation initiated.

Build platform BP-316L-89 installed and leveled.

Process parameters loaded: SLM-316L-Washer-Profile-v2.3

09:30 - Chamber sealing verified. Inert atmosphere establishment began.

Argon gas usage was determined from the standard procedure: initial chamber purge of 700 liters plus 54 liters per component for the 33 washers in this batch. The total argon consumption was calculated as 3.08 kg.

10:00 - SLM process started.

Machine status: Nominal power with laser on: 5.5 kW

Machine status: Nominal power with laser off: 3.5 kW

Processing Log Entries:

Time	Activity	Operator	Notes
10:00	Build started	M. Chen	Layer 1 initiated
12:30	Mid-process inspection	M. Chen	All systems nominal
15:45	Powder replenishment	M. Chen	Added 0.5 kg from recovery system
23:18	Build completed	M. Chen	Total processing time: 13.38 hours

The SLM process electricity consumption was calculated based on the processing time of 13.38 hours and the machine’s power profile during operation, resulting in 64.92 kWh of low-voltage electricity usage.

23:30 - Process completed. Chamber venting initiated.

4.0 Post-Processing and Recovery

23:45 - Build platform removal and part extraction.

00:15 - Powder recovery process initiated.

Material Flow	Quantity	Destination
316L Powder Reused in SLM Process	2.94 kg	SLM Powder Handling System
316L Powder Returned for Remelting	0.15 kg	Water Atomization Department
Recovered Process Water	16.4 kg	Water Treatment System

00:45 - Support structure removal and initial cleaning completed.

01:15 - Waste materials segregated and logged.

Waste Type	Quantity	Disposition
Solid Waste from Water Atomization	0.41 kg	Landfill (Hazardous Waste Facility)
Non-recyclable 316L Powder from SLM	0.01 kg	Landfill (Hazardous Waste Facility)

5.0 Quality Control and Final Inspection

01:30 - Visual inspection of all components.

02:00 - Dimensional verification sampling (10% of batch).

Sample	OD (mm)	ID (mm)	Thickness (mm)
1	19.98	9.01	1.49
2	20.01	8.99	1.51
3	19.99	9.02	1.50
All within specification: OD 20.0±0.1mm, ID 9.0±0.1mm, Thickness 1.5±0.1mm			

02:30 - Final product weighing and count verification.

Product	Quantity	Total Mass
316L Stainless Steel Flat Washers	33 units	0.61 kg

03:00 - Documentation review completed. All quality checks passed.

6.0 Equipment and Utility Data

SLM Machine Performance Data:

Parameter	Value	Notes
Total Processing Time	13.38 hours	From start to completion
Laser Operating Efficiency	94.2%	Within normal range (92-96%)
Chamber Temperature Stability	±1.5°C	Specification: ±2.0°C
Oxygen Level in Chamber	<100 ppm	Required: <1000 ppm

Utility Consumption Summary:

Electricity meter readings for the production area:

- Start of shift: 284,567.3 kWh
- End of shift: 284,634.8 kWh

Note: Total consumption includes lighting, ventilation, and auxiliary equipment in addition to process equipment.

Gas System Data:

Argon supply pressure maintained at 4.5-5.0 bar throughout process.
No leaks detected during pressure decay test.

7.0 Batch Completion and Sign-off

03:15 - All production activities completed.

03:30 - Final documentation review.

Batch Yield Calculation:

Target: 33 units

Produced: 33 units

Yield: 100%

Comparative Performance Data:

Previous batch (SLM-316L-WSH-231008): Processing time 13.45 hours, yield 97% (32/33 units)

Department benchmark: Typical yield 95-98%, processing time 13.2-13.6 hours

Disposition:

 BATCH APPROVED FOR SHIPMENT

Signatures:

Role	Name	Signature	Date
Lead Operator	Michael Chen		October 16, 2023
Quality Inspector	Sarah Rodriguez		October 16, 2023
Production Supervisor	John Doe		October 16, 2023

8.0 Notes and Observations

- No equipment malfunctions or process deviations recorded
- Powder handling efficiency improved compared to previous batch (recovery rate increased by 2%)
- All safety protocols followed throughout the process
- Next preventive maintenance scheduled for November 15, 2023

End of Batch Production Record